10:- E Commerce Website:-

```
<!DOCTYPE html>

<html>

<head>

<title>SmartShop</title>

</head>

<body style="text-align:center; font-family:Arial;">


<h2>SmartShop 🛍️</h2>

<p>Select a product:</p>


<button onclick="add('Shoes')">Shoes</button>

<button onclick="add('Watch')">Watch</button>

<button onclick="add('T-Shirt')">T-Shirt</button>

<button onclick="add('Headphones')">Headphones</button>


<h3>Recommended For You:</h3>

<p id="rec">No recommendations yet.</p>


<button onclick="clearRec()">Clear</button>


<script>

function add(item) {

   localStorage.setItem("product", item);
```

```
    show();
}

function show() {
  let product = localStorage.getItem("product");
  let rec = document.getElementById("rec");

  if (product) {
    rec.innerHTML = "You viewed: " + product + "<br>You may like similar items!";
  } else {
    rec.innerHTML = "No recommendations yet.";
  }
}

function clearRec() {
    localStorage.removeItem("product");
    show();
}

window.onload = show;
</script>

</body>
</html>
```

9:- E Commerce Website:-

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Simple Registration Form</title>

<style>
  body {
    text-align: center;
    font-family: Arial;
    background-color: #eef2f3;
    padding-top: 40px;
  }
  input {
    margin: 8px;
    padding: 8px;
  }
  .error {
    color: red;
    font-size: 14px;
  }
  .success {
```

```
        color: green;
        font-size: 16px;
        margin-top: 10px;
    }
</style>

</head>
<body>

<h2>Register Here</h2>

<form id="form">
    <input type="text" id="user" placeholder="Username"><br>
    <span id="userErr" class="error"></span><br>

    <input type="password" id="pass" placeholder="Password"><br>
    <span id="passErr" class="error"></span><br>

    <button type="submit">Submit</button>
</form>

<div id="message" class="success"></div>

<script>
    const form = document.getElementById("form");
    const user = document.getElementById("user");
```

```
const pass = document.getElementById("pass");
const userErr = document.getElementById("userErr");
const passErr = document.getElementById("passErr");
const message = document.getElementById("message");

form.addEventListener("submit", function (e) {
  e.preventDefault();

  userErr.textContent = "";
  passErr.textContent = "";
  message.textContent = "";

  let valid = true;

  if (user.value.length < 5) {
    userErr.textContent = "Username must be at least 5 characters.";
    valid = false;
  }

  if (pass.value.length < 6) {
    passErr.textContent = "Password must be at least 6 characters.";
    valid = false;
  }

  if (valid) {
    message.textContent = "✅ Registration Successful!";
```

```
            form.reset();
        }
    });
</script>

</body>
</html>
```

8:-Movie Recommendation System

```
<!DOCTYPE html>
<html>
<head>
<title>Simple Movie Recommendation</title>
<style>
body { text-align:center; font-family:Arial; background:#f0f0f0; padding-top:50px; }
.box { background:white; padding:20px; display:inline-block; border:1px solid #ccc; }
select, button { margin:10px; padding:8px; }
#result { margin-top:15px; font-size:18px; font-weight:bold; }
</style>
</head>

<body>

<div class="box">
<h2>Movie Recommendation</h2>

<select id="user">
<option>Alice</option>
<option>Bob</option>
<option>Charlie</option>
<option>David</option>
</select><br>
```

```
<button onclick="getRec()">Get Recommendation</button>


<p id="result"></p>
</div>
<script>
const ratings = {
  Alice:   { Matrix:5, Titanic:3, Inception:4, Avatar:0 },
  Bob:     { Matrix:4, Titanic:5, Inception:0, Avatar:3 },
  Charlie: { Matrix:0, Titanic:4, Inception:5, Avatar:2 },
  David:   { Matrix:5, Titanic:0, Inception:4, Avatar:5 }
};
function getRec() {
  let user = document.getElementById("user").value;
  let result = document.getElementById("result");


  for (let movie in ratings[user]) {
    if (ratings[user][movie] === 0) {
      result.innerHTML = "Suggested Movie: " + movie;
      return;
    }
  }
  result.innerHTML = "No movies left!";
}
</script>
</body>
</html>
```

7:-NLP Text to speech

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<title>Simple NLP Text-to-Speech</title>


<!-- ✅ NLP Library -->

<script
src="https://cdn.jsdelivr.net/npm/compromise@13.11.3/builds/compromise.min.js"></script>


<style>

body { text-align:center; font-family:Arial; margin-top:40px; }

textarea { width:300px; height:100px; }

</style>

</head>

<body>


<h2>NLP Text-to-Speech</h2>


<textarea id="text" placeholder="Type text here..."></textarea><br><br>

<button onclick="processText()">Speak</button>


<p id="output"></p>
```

```
<script>
function processText() {
    let input = document.getElementById("text").value;

    if (!input.trim()) return;

    // ✅ NLP Processing
    let doc = nlp(input);
    let nouns = doc.nouns().out("array");



    // ✅ Text-to-Speech
    let speech = new SpeechSynthesisUtterance(input);
    speechSynthesis.speak(speech);
}
</script>

</body>
</html>
```

6:-Spam Detection

```
<!DOCTYPE html>
<html>
<head>
<title>Simple Spam Checker</title>
<script
src="https://cdn.jsdelivr.net/npm/compromise@13.11.3/builds/compromise.min.js"></script>
<style>
body { text-align:center; font-family:Arial; margin-top:50px; }
.result { font-size:20px; font-weight:bold; margin-top:15px; }
</style>
</head>

<body>

<h3>Spam Checker (NLP)</h3>

<textarea id="msg" placeholder="Type message..." rows="4" cols="30"></textarea><br><br>

<button onclick="check()">Check Spam</button>

<p id="out" class="result"></p>

<script>
function check() {
```

```
  let text = document.getElementById("msg").value.toLowerCase();
  let out = document.getElementById("out");

  if(text === "") {
    out.innerHTML = "Please enter a message!";
    return;
  }

  let doc = nlp(text);
  let words = doc.words().out("array");

  let spam = ["free","win","money","lottery","prize","offer"];
  let found = words.filter(w => spam.includes(w));

  if(found.length > 0) {
    out.innerHTML = "🚫 SPAM Message!";
    out.style.color = "red";
  } else {
    out.innerHTML = "✅ Not Spam";
    out.style.color = "green";
  }
}
</script>

</body>
</html>
```

5:-Sentiment Analysis

```
<!DOCTYPE html>

<html>

<head>

<title>Simple Sentiment Check</title>

<style>

  body {

    text-align: center;

    font-family: Arial;

    margin-top: 50px;

  }

  textarea {

    width: 300px;

    height: 80px;

    font-size: 16px;

  }

</style>

</head>

<body>


<h2>Sentiment Checker</h2>

<textarea id="text" placeholder="Write something..."></textarea><br><br>

<button onclick="checkSentiment()">Check</button>
```

```
<h3 id="output"></h3>

<script>
function checkSentiment() {
   let input = document.getElementById("text").value.toLowerCase();

   let positive = ["good", "happy", "love"];
   let negative = ["bad", "sad", "hate"];

   let score = 0;

   positive.forEach(word => {
      if (input.includes(word)) score++;
   });

   negative.forEach(word => {
      if (input.inclucdes(word)) score--;
   });

   if (score > 0) {
      document.getElementById("output").innerHTML = "Positive 😊";
   }
   else if (score < 0) {
      document.getElementById("output").innerHTML = "Negative 😔";
   }
   else {
```

```
        document.getElementById("output").innerHTML = "Neutral 😐";
    }
}
</script>

</body>
</html>
```

2:- TensorFlow face detection

```
<!DOCTYPE html>
<html>
<body style="text-align:center; font-family:Arial; margin-top:30px">

<h2>Simple Face Detection</h2>

<input type="file" id="fileInput" accept="image/*"><br><br>
<canvas id="canvas"></canvas>
<p id="msg"></p>

<!-- TensorFlow & BlazeFace -->
<script src="https://cdn.jsdelivr.net/npm/@tensorflow/tfjs"></script>
<script src="https://cdn.jsdelivr.net/npm/@tensorflow-models/blazeface"></script>

<script>
let model;
let canvas = document.getElementById("canvas");
let ctx = canvas.getContext("2d");
let msg = document.getElementById("msg");

// Load the model
blazeface.load().then(m => {
  model = m;
  msg.innerHTML = "✅ Model Loaded! Select an image.";
```

```
});

// When image uploaded
fileInput.onchange = async () => {
  if (!model) return;

  let img = new Image();
  img.crossOrigin = "anonymous"; // ✅ Fix: allow canvas drawing
  img.src = URL.createObjectURL(fileInput.files[0]);

  img.onload = async () => {
    canvas.width = img.width;
    canvas.height = img.height;
    ctx.drawImage(img, 0, 0);

    const faces = await model.estimateFaces(img);

    if (faces.length === 0) {
      msg.innerHTML = "❌ No Face Found";
      return;
    }

    msg.innerHTML = "✅ Face Detected: " + faces.length;

    ctx.strokeStyle = "red";
    ctx.lineWidth = 3;
```

```
        faces.forEach(face => {
            const [x1, y1] = face.topLeft;
            const [x2, y2] = face.bottomRight;
            ctx.strokeRect(x1, y1, x2 - x1, y2 - y1);
        });
    };
};
</script>

</body>
</html>
```

}

1:- Student Details form

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<title>Student Details Form</title>


<style>

  body {

    font-family: Arial, sans-serif;

    background-color: #eef7ff;

    text-align: center;

    padding-top: 40px;

  }


  .form-box {

    background: #fff;

    width: 380px;

    margin: auto;

    padding: 20px;

    border-radius: 10px;

    box-shadow: 0px 0px 10px gray;

    text-align: left;

  }
```

```css
label {
    font-size: 18px;
    font-weight: bold;
}

input, select, textarea {
    width: 100%;
    padding: 8px;
    margin: 8px 0 16px;
    border-radius: 5px;
    border: 1px solid #555;
}

button {
    width: 100%;
    padding: 10px;
    font-size: 18px;
    border: none;
    background-color: #0066cc;
    color: white;
    border-radius: 5px;
    cursor: pointer;
}

button:hover {
    background-color: #004c99;
```

```
    }
</style>
</head>

<body>

<h2>Student Details Form</h2>

<div class="form-box">
    <label>Name:</label>
    <input type="text" placeholder="Enter student name">

    <label>Roll Number:</label>
    <input type="number" placeholder="Enter roll number">

    <label>Department:</label>
    <input type="text" placeholder="Enter department">

    <label>City:</label>
    <input type="text" placeholder="Enter city">

    <label>Address:</label>
    <textarea placeholder="Enter address"></textarea>

    <label>Age:</label>
    <input type="number" placeholder="Enter age">
```

```
    <label>Gender:</label>
    <select>
      <option>Select Gender</option>
      <option>Male</option>
      <option>Female</option>
      <option>Other</option>
    </select>

    <label>Email:</label>
    <input type="email" placeholder="Enter email">

    <button>Submit</button>
    <button type="reset">Reset</button>
  </div>

</body>
</html>
```

3,4:- API

```
const express = require('express');
const bodyParser = require('body-parser');
const mongoose = require('mongoose');

const app = express();
app.use(bodyParser.json());
const PORT = 5000;

// MongoDB connection
mongoose.connect("mongodb://localhost:27017/DEMODB")
.then(() => console.log('Connected to MongoDB'))
.catch(err => console.error('Could not connect to MongoDB...', err));

// Schema
const userSchema = new mongoose.Schema({
uname: { type: String, unique: true, required: true },
password: { type: String, required: true }
});

const User = mongoose.model('Users', userSchema);

// Create user (Register)
```

```
app.post('/users', async (req, res) => {
try {
const { uname, password } = req.body;

// Check if user already exists
const existingUser = await User.findOne({ uname });
if (existingUser) {
return res.status(400).json({ error: 'Username already exists' });
}

// Hash password before saving


const newUser = new User({ uname, password});
const savedUser = await newUser.save();

// Respond without password
res.status(201).json({ _id: savedUser._id, uname: savedUser.uname });
} catch (err) {
res.status(500).json({ error: err.message });
}
});
// Login route
app.post('/login', async (req, res) => {
try {
const { uname, password } = req.body;
```

```
// Find user by uname and match password directly
const user = await User.findOne({ uname, password });
if (!user) {
return res.status(400).json({ error: 'Invalid username or password' });
}

// Successful login
res.status(200).json({ message: 'Login successful', uname: user.uname });
} catch (err) {
res.status(500).json({ error: err.message });
}
});


// Get all users (for testing)
app.get('/users', async (req, res) => {
try {
const users = await User.find({}, { password: 0 }); // Exclude passwords
res.json(users);
} catch (err) {
res.status(500).json({ error: err.message });
}
});

app.listen(PORT, () => {
console.log(Server is Running on http://localhost:${PORT});
```

```
})
```